2
Judea Pearl and the Ladder of Causation

In the last chapter, we discussed why association is not sufficient to draw causal conclusions. We talked about **interventions** and **counterfactuals** as tools that allow us to perform causal inference based on observational data. Now, it's time to give it a bit more structure.

In this chapter, we're going to introduce the concept of **the Ladder of Causation**. We'll discuss associations, interventions, and counterfactuals from theoretical and mathematical standpoints. Finally, we'll implement a couple of **structural causal models** in Python to solidify our understanding of the three aforementioned concepts. By the end of this chapter, you should have a firm grasp of the differences between associations, interventions, and counterfactuals. This knowledge will be a foundation of many of the ideas that we'll discuss further in the book and allow us to understand the mechanics of more sophisticated methods that we'll introduce in *Part 2*, *Causal Inference*, and *Part 3*, *Causal Discovery*.

In this chapter, we will cover the following topics:

- The concept of the Ladder of Causation
- Conceptual, mathematical, and practical differences between associations, interventions, and counterfactuals

From associations to logic and imagination – the Ladder of Causation

In this section, we'll introduce the concept of *the Ladder of Causation* and summarize its building blocks. *Figure 2.1* presents a symbolic representation of the Ladder of Causation. The higher the rung, the more sophisticated our capabilities become, but let's start from the beginning:

Figure 2.1 – The Ladder of Causation. Image by the author, based on a picture by Laurie Shaw (`https://www.pexels.com/photo/brown-wooden-door-frame-804394/`)

The Ladder of Causation, introduced by Judea Pearl (Pearl, Mackenzie, 2019), is a helpful metaphor for understanding distinct levels of relationships between variables – from simple associations to counterfactual reasoning. Pearl's ladder has three rungs. Each rung is related to different activity and offers answers to different types of causal questions. Each rung comes with a distinct set of mathematical tools.

Judea Pearl

Judea Pearl is an Israeli-American researcher and computer scientist, who devoted a large part of his career to researching causality. His original and insightful work has been recognized by **Association for Computing Machinery** (**ACM**), who awarded him with the Turing Award – considered by many the equivalent of the Nobel Prize in computer science. *The Ladder of Causation* was introduced in Pearl's popular book on causality, *The Book of Why* (Pearl, Mackenzie, 2019).

Rung one of the ladder represents **association**. The activity that is related to this level is *observing*. Using association, we can answer questions about how seeing one thing changes our beliefs about another thing – for instance, how observing a successful space launch by SpaceX changes our belief that SpaceX stock price will go up.

Rung two represents **intervention**. Remember the babies from the previous chapter? The action related to rung two is *doing* or *intervening*. Just like babies throwing their toys around to learn about the laws of physics, we can intervene on one variable to check how it influences some other variable. Interventions can help us answer questions about what will happen to one thing if we change another thing – for instance, if I go to bed earlier, will I have more energy the following morning?

Rung three represents **counterfactual reasoning**. Activities associated with rung three are *imagining* and *understanding*. Counterfactuals are useful to answer questions about what would have happened if we had done something differently. For instance, would I have made it to the office on time if I took the train rather than the car?

Table 2.1 summarizes the three rungs of *the Ladder of Causation*:

Rung	**Action**	**Question**
Association (1)	*Observing*	How does observing X change my belief in Y?
Intervention (2)	*Doing*	What will happen to Y if I do X?
Counterfactual (3)	*Imagining*	If I had done X, what would Y be?

Table 2.1 – A summary of the three rungs of the Ladder of Causation

To cement our intuitions, let's see an example of each of the rungs.

Imagine that you're a doctor and you consider prescribing drug *D* to one of your patients. First, you might recall hearing other doctors saying that *D* helped their patients. It seems that in the sample of doctors you heard talking about *D*, there is an association between their patients taking the drug and getting better. That's rung one. We are skeptical about the rung one evidence because it might just be the case that these doctors only treated patients with certain characteristics (maybe just mild cases or only patients of a certain age). To overcome the limitation of rung one, you decide to read articles based on randomized clinical trials.

Randomized controlled trials

Randomized controlled trials (**RCTs**), sometimes referred to as *randomized experiments*, are often considered the gold standard for causal inference. The key idea behind RCTs is **randomization**. We randomly assign subjects in an experiment to treatment and control groups, which helps us achieve *deconfounding*. There are many possible RCT designs. For an introduction and discussion, check out Matthews (2006). We also briefly discuss RCTs in *Chapter 12*.

These trials were based on interventions (rung two) and – assuming that they were properly designed – they can be used to determine the relative efficacy of the treatment. Unfortunately, they cannot tell us whether a patient would be better off if they had taken the treatment earlier, or which of two available treatments with similar relative efficacy would have worked better for this particular patient. To answer this type of question, we need rung three.

Now, let's take a closer look at each of the rungs and their respective mathematical apparatus.

Associations

In this section, we'll demonstrate how to quantify **associational relationships** using **conditional probability**. Then, we'll briefly introduce structural causal models. Finally, we'll implement conditional probability queries using Python.

We already learned a lot about associations. We know that associations are related to *observing* and that they allow us to generate predictions. Let's take a look at mathematical tools that will allow us to talk about associations in a more formal way.

We can view the mathematics of rung one from a couple of angles. In this section, we'll focus on the perspective of **conditional probability**.

> **Conditional probability**
>
> Conditional probability is the probability of one event, given that another event has occurred. A mathematical symbol that we use to express conditional probability is **|** (known as a *pipe* or *vertical bar*). We read $P(X|Y)$ as a *probability of X given Y*. This notation is a bit simplified (or abused if you will). What we usually mean by $P(X|Y)$ is $P(X = x|Y = y)$, the probability that the variable X takes the value x, given that the variable Y takes the value y. This notation can also be extended to continuous cases, where we want to work with probability densities – for example, $P(0 < X < 0.25|Y > 0.5)$.

Imagine that you run an internet bookstore. What is the probability that a person will buy book A, given that they bought book B? This question can be answered using the following conditional probability query:

$P(book\ A|book\ B)$

Note that the preceding formula does not give us any information on the causal relationship between both events. We don't know whether buying book A *caused* the customer to buy book B, buying book B caused them to buy book A, or there is another (unobserved) event that caused both. We only get information about *non-causal association* between these events. To see this clearly, we will implement our bookstore example in Python, but before we start, we'll briefly introduce one more important concept.

Structural causal models (**SCMs**) are a simple yet powerful tool to encode causal relationships between variables. You might be surprised that we are discussing a causal model in the section on association. Didn't we just say that association is usually not enough to address causal questions? That's true. The reason why we're introducing an SCM now is that we will use it as our *data-generating process*. After generating the data, we will pretend to forget what the SCM was. This way, we'll mimic a frequent real-world scenario where the true data-generating process is unknown, and the only thing we have is observational data.

Let's take a small detour from our bookstore example and take a look at *Figure 2.2*:

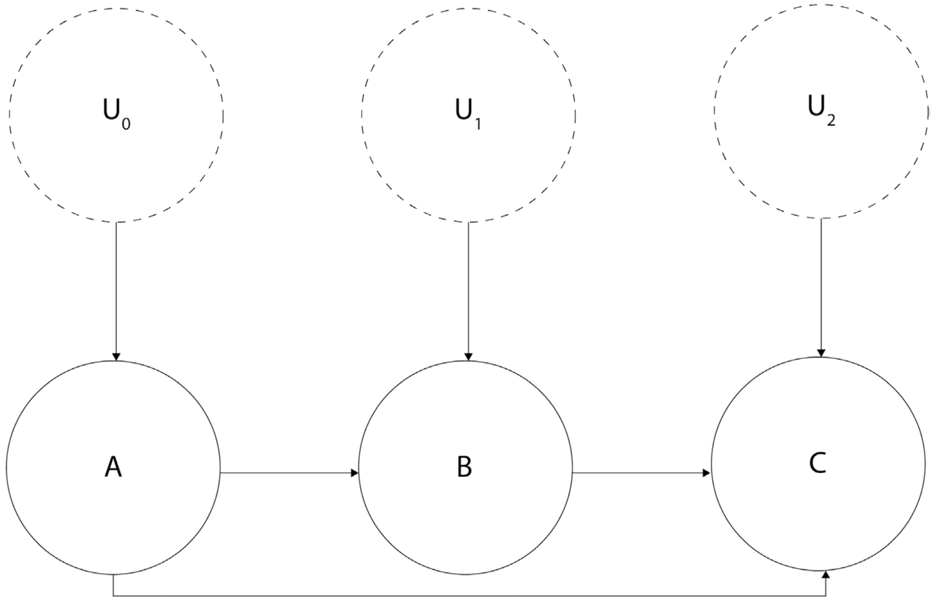

Figure 2.2 – A graphical representation of a structural causal model

Circles or *nodes* in the preceding figure represent variables. Lines with arrows or *edges* represent relationships between variables.

As you can see, there are two types of variables (marked with dashed versus regular lines). Arrows at the end of the lines represent the *direction* of the relationship.

Nodes **A**, **B**, and **C** are marked with solid lines. They represent the observed variables in our model. We call this type of variable **endogenous**. Endogenous variables are always children of at least one other variable in a model.

The other type of nodes (U_X nodes) are marked with dashed lines. We call these variables **exogenous**, and they are represented by *root* nodes in the graph (they are not descendants of any other variable; Pearl, Glymour, and Jewell, 2016). Exogenous variables are also called **noise variables**.

Noise variables

Note that most causal inference and causal discovery methods require that noise variables are uncorrelated with each other (otherwise, they become unobserved confounders). This is one of the major difficulties in real-world causal inference, as sometimes, it's very hard to be sure that we have met this assumption.

An SCM can be represented graphically (as in *Figure 2.2*) or as a series of equations. These two representations have different properties and might require different assumptions (we'll leave this complexity out for now), but they refer to the same object – a **data-generating process**.

Let's return to the SCM from *Figure 2.2*. We'll define the functional relationships in this model in the following way:

$$A := f_A(U_0)$$

$$B := f_B(A, U_1)$$

$$C := f_C(A, B, U_2)$$

A, B, C, and U_X represent the nodes in *Figure 2.1*, and := is an **assignment operator**, also known as a **walrus operator**. We use it here to emphasize that the relationship that we're describing is *directional* (or asymmetric), as opposed to the regular equal sign that suggests a symmetric relation. Finally, f_A, f_B, f_C represent arbitrary functions (they can be as simple as a summation or as complex as you want). This is all we need to know about SCMs at this stage. We will learn more about them in *Chapter 3*.

Equipped with a basic understanding of SCMs, we are now ready to jump to our coding exercise in the next section.

Let's practice!

For this exercise, let's recall our bookstore example from the beginning of the section.

First, let's define an SCM that can generate data with a non-zero probability of buying book A, given we bought book B. There are many possible SCMs that could generate such data. *Figure 2.3* presents the model we have chosen for this section:

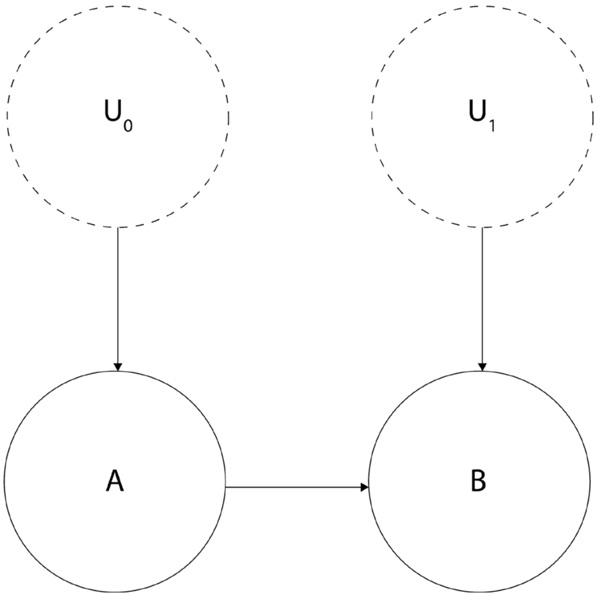

Figure 2.3 – A graphical model representing the bookstore example

To precisely define causal relations that drive our SCM, let's write a set of equations:

$$U_0 \sim U(0, 1)$$

$$U_1 \sim N(0, 1)$$

$$A := 1_{\{U_0 > .61\}}$$

$$B := 1_{\{(A + .5 * U_1) > .2\}}$$

In the preceding formulas, U_0 is a continuous random variable uniformly distributed between 0 and 1. U_1 is a normally distributed random variable, with a mean value of 0 and a standard deviation of 1. A and B are binary variables, and $1_{\{f\}}$ is an indicator function.

The indicator function

The notation for the indicator function might look complicated, but the idea behind it is very simple. The indicator function returns 1 when the condition in the curly braces is met and returns 0 otherwise. For instance, let's take the following function:

$$X = 1_{\{Z>0\}}$$

If $Z > 0$ then $X = 1$, otherwise $X = 0$.

Now, let's recreate this SCM in code. You can find the code for this exercise in the notebook for this chapter `https://bit.ly/causal-ntbk-02`:

1. First, let's import the necessary libraries:

```
import numpy as np
from scipy import stats
```

2. Next, let's define the SCM. We will use the object-oriented approach for this purpose, although you might want to choose other ways for yourself, which is perfectly fine:

```
class BookSCM:

    def __init__(self, random_seed=None):
        self.random_seed = random_seed
        self.u_0 = stats.uniform()
        self.u_1 = stats.norm()

    def sample(self, sample_size=100):
        """Samples from the SCM"""
        if self.random_seed:
            np.random.seed(self.random_seed)

        u_0 = self.u_0.rvs(sample_size)
```

```
        u_1 = self.u_1.rvs(sample_size)
        a = u_0 > .61
        b = (a + .5 * u_1) > .2

        return a, b
```

Let's unpack this code. In the `__init__()` method of our `BookSCM`, we define the distributions for U_0 and U_1 and set a random seed for reproducibility; the `.sample()` method samples from U_0 and U_1, computes values for A and B (according to the formulas specified previously), and returns them.

Great! We're now ready to generate some data and quantify an association between the variables using *conditional probability*:

1. First, let's instantiate our SCM and set the random seed to `45`:

```
scm = BookSCM(random_seed=45)
```

2. Next, let's sample `100` samples from it:

```
buy_book_a, buy_book_b = scm.sample(100)
```

Let's check whether the shapes are as expected:

```
buy_book_a.shape, buy_book_b.shape
```

The output is as follows:

```
((100,), (100,))
```

The shapes are correct. We generated the data, and we're now ready to answer the question that we posed at the beginning of this section – what is the probability that a person will buy book A, given that they bought book B?

3. Let's compute the $P(book\ A|book\ B)$ conditional probability to answer our question:

```
proba_book_a_given_book_b = buy_book_a[buy_book_b].sum() / buy_
book_a[buy_book_b].shape[0]
print(f'Probability of buying book A given B: {proba_book_a_
given_book_b:0.3f}')
```

This returns the following result:

```
Probability of buying book A given B: 0.638
```

As we can see, the probability of buying book A, given we bought book B, is 63.8%. This indicates a positive relationship between both variables (if there was no association between them, we would expect the result to be 50%). These results inform us that we can make meaningful predictions using observational data alone. This ability is the essence of most contemporary (supervised) machine learning models.

Let's summarize. Associations are useful. They allow us to generate meaningful predictions of potentially high practical significance in the absence of knowledge of the data-generating process. We used an SCM to generate hypothetical data for our bookstore example and estimated the strength of association between book A and book B sales, using a conditional probability query. Conditional probability allowed us to draw conclusions in the absence of knowledge of the true data-generating process, based on the observational data alone (note that although we knew the true SCM, we did not use any knowledge about it when computing the conditional probability query; we virtually *forgot* anything about the SCM before generating the predictions). That said, associations only allow us to answer rung one questions.

Let's climb to the second rung of *the Ladder of Causation* to see how to go beyond some of these limitations.

What are interventions?

In this section, we'll summarize what we've learned about interventions so far and introduce mathematical tools to describe them. Finally, we'll use our newly acquired knowledge to implement an intervention example in Python.

The idea of intervention is very simple. We change one thing in the world and observe whether and how this change affects another thing in the world. This is the essence of scientific experiments. To describe interventions mathematically, we use a special *do*-operator. We usually express it in mathematical notation in the following way:

$$P(Y = 1 | do(X = 0))$$

The preceding formula states that the probability of $Y = 1$, given that we set X to 0. The fact that we need to change X's value is critical here, and it highlights the inherent difference between *intervening* and *conditioning* (conditioning is the operation that we used to obtain conditional probabilities in the previous section). Conditioning only modifies our *view* of the data, while intervening affects the distribution by *actively* setting one (or more) variable(s) to a *fixed value* (or a distribution). This is very important – intervention *changes* the system, but conditioning *does not*. You might ask, what does it mean that *intervention changes the system*? Great question!

The graph saga – parents, children, and more

When we talk about graphs, we often use terms such as *parents*, *children*, *descendants*, and *ancestors*. To make sure that you can understand the next subsection clearly, we'll give you a brief overview of these terms here.

We say that the node X is a *parent* of the node Y and that Y is a *child* of X when there's a direct arrow from X to Y. If there's also an arrow from Y to Z, we say that Z is a grandchild of X and that X is a grandparent of Z. Every child of X, all its children and their children, their children's children, and so on are descendants of X, which is their *ancestor*. For a more formal explanation, check out *Chapter 4*.

Changing the world

When we intervene in a system and fix a value or alter the distribution of some variable – let's call it X – one of three things can happen:

- The change in X will influence the values of its descendants (assuming X has descendants and excluding special cases where X's influence is canceled – for example, $f(x) = x - x$)
- X will become independent of its ancestors (assuming that X has ancestors)
- Both situations will take place (assuming that X has descendants and ascendants, excluding special cases)

Note that none of these would happen if we *conditioned* on X, because conditioning *does not change* the value of any of the variables – it *does not* change the system.

Let's translate interventions into code. We will use the following SCM for this purpose:

$U_0 \sim N(0, 1)$

$U_1 \sim N(0, 1)$

$A := U_0$

$B := 5A + U_1$

The graphical representation of this model is identical to the one in *Figure 2.3*. Its functional assignments are different though, and – importantly – we set A and B to be continuous variables (as opposed to the model in the previous section, where A and B were binary; note that this is a new example, and the only thing it shares with the bookstore example is the structure of the graph):

1. First, we'll define the sample size for our experiment and set a random seed for reproducibility:

```
SAMPLE_SIZE = 100
np.random.seed(45)
```

2. Next, let's build our SCM. We will also compute the correlation coefficient between A and B and print out a couple of statistics:

```
u_0 = np.random.randn(SAMPLE_SIZE)
u_1 = np.random.randn(SAMPLE_SIZE)
a = u_0
b = 5 * a + u_1

r, p = stats.pearsonr(a, b)
print(f'Mean of B before any intervention: {b.mean():.3f}')
print(f'Variance of B before any intervention: {b.var():.3f}')
print(f'Correlation between A and B:\nr = {r:.3f}; p =
{p:.3f}\n')
```

We obtain the following result:

```
Mean of B before any intervention: -0.620
Variance of B before any intervention: 22.667
Correlation between A and B:
r = 0.978; p = 0.000
```

As we can see, the correlation between values of A and B is very high ($r = .978; p < .001$). It's not surprising, given that B is a simple linear function of A. The mean of B is slightly below zero, and the variance is around `22`.

3. Now, let's intervene on A by fixing its value at `1.5`:

```
a = np.array([1.5] * SAMPLE_SIZE)
b = 5 * a + u_1
```

We said that an intervention *changes the system*. If that's true, the statistics for B should change as a result of our intervention. Let's check it out:

```
print(f'Mean of B after the intervention on A: {b.mean():.3f}')
print(f'Variance of B after the intervention on A:
{b.var():.3f}\n')
```

The result is as follows:

```
Mean of B after the intervention on A: 7.686
Variance of B after the intervention on A: 0.995
```

Both the mean and variance have changed. The new mean of B is significantly greater than the previous one. This is because the value of our intervention on A (1.5) is much bigger than what we'd expect from the original distribution of A (centered at 0). At the same time, the variance has shrunk. This is because A became constant, and the only remaining variability in B comes from its stochastic parent, U_1.

What would happen if we intervened on B instead? Let's see and print out a couple of statistics:

```
a = u_0
b = np.random.randn(SAMPLE_SIZE)

r, p = stats.pearsonr(a, b)

print(f'Mean of B after the intervention on B: {b.mean():.3f}')
print(f'Variance of B after the intervention on B: {b.var():.3f}')
print(f'Correlation between A and B after intervening on B:\nr =
{r:.3f}; p = {p:.3f}\n')
```

This results in the following output:

```
Mean of B after the intervention on B: 0.186
Variance of B after the intervention on B: 0.995
Correlation between A and B after intervening on B:
r = -0.023; p = 0.821
```

Note that the correlation between A and B dropped to almost zero ($r = -.023$), and the corresponding p-value indicates a lack of significance ($p = .821$). This indicates that after the intervention, A and B became (linearly) independent. This result suggests that there is no causal link from B to A. At the same time, previous results demonstrated that intervening on A changes B, indicating that there is a causal link from A to B (we'll address what a *causal link* means more systematically in *Chapter 4*).

Before we conclude this section, we need to mention one more thing.

Correlation and causation

You might have heard the phrase that *correlation is not causation*. That's approximately true.

How about the opposite statement? Is *causation correlation*?

Let's see.

Figure 2.4 presents the data generated according to the following set of structural equations:

$X := \mathcal{U}(-2, 2)$

$Y := X^2 + 0.2 \times \mathcal{N}(0, 1)$

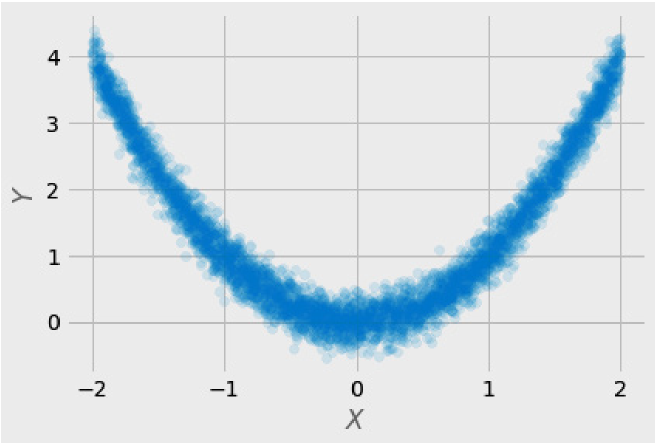

Figure 2.4 – A scatter plot of the data from a causal data-generating process

Although from the structural point of view, there's a clear causal link between X and Y, the correlation coefficient for this dataset is essentially equal to 0 (you can experiment with this data in the notebook: `https://bit.ly/causal-ntbk-02`). The reason for this is that the relationship between X and Y is not *monotonic*, and popular correlation metrics such as Pearson's r or Spearman's *rho* cannot capture non-monotonic relationships. This leads us to an important realization that a lack of traditional correlation does not imply independence between variables.

A number of tools for more general independence testing exist. For instance, information-theoretic metrics such as the **maximal information coefficient** (**MIC**) (Reshef et al., 2011; Reshef et al., 2015; Murphy, 2022, pp. 217–219) work for non-linear, non-monotonic data out of the box. The same goes for the **Hilbert-Schmidt independence criterion** (**HSIC**) (Gretton et al., 2007) and a number of other metrics.

Another scenario where you might see no correlation although causation is *present* is when your sampling does not cover the entire support of relevant variables. Take a look at *Figure 2.5*.

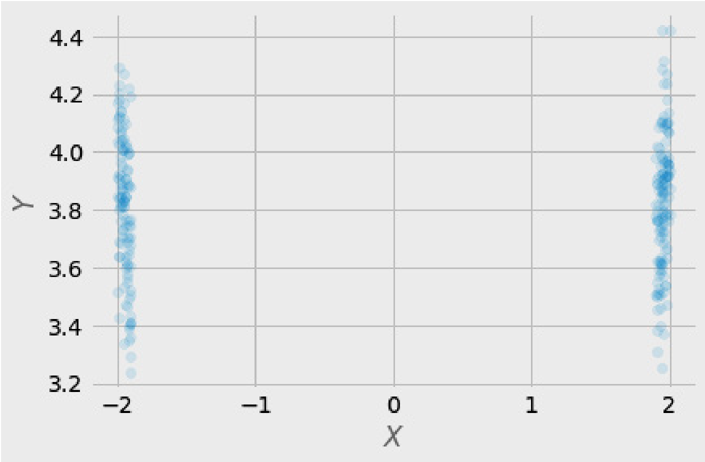

Figure 2.5 – A scatterplot of data with selective sampling.

This data is a result of exactly the same process as the data presented in *Figure 2.4*. The only difference is that we sampled according to the following condition:

$$X_{sample} = 1.9 < X < -1.9$$

It's virtually impossible to estimate the true relationship between X and Y from this data, even with sophisticated tools.

Situations such as these can happen in real life. Everything from faulty sensors to selection bias in medical studies can remove a substantial amount of information from observational data and push us toward wrong conclusions.

In this section, we looked into interventions in greater detail. We introduced the *do*-operator and discussed how it differs from conditioning. Finally, we implemented a simple SCM in Python and demonstrated how interventions on one variable affect the distribution of other variables.

If all of this makes you a little dizzy, don't worry. For the vast majority of people coming from a statistical background, conversion to causal thinking takes time. This is because causal thinking requires breaking certain habits we all pick up when we learn statistical thinking. One such habit is thinking in terms of variables as basic entities rather than in terms of the process that generated these variables.

Ready for more? Let's step up to rung three – the world of counterfactuals!

What are counterfactuals?

Have you ever wondered where you would be today if you had chosen something different in your life? Moved to another city 10 years ago? Studied art? Dated another person? Taken a motorcycle trip in Hawaii? Answering these types of questions requires us to create alternative worlds, worlds that we have never observed. If you've ever tried doing this for yourself, you already know intuitively what counterfactuals are.

Let's try to structure this intuition. We can think about counterfactuals as a minimal modification to a system (Pearl, Glymour, and Jewell, 2016). In this sense, they are similar to interventions. Nonetheless, there is a fundamental difference between the two.

Counterfactuals can be thought of as *hypothetical* or *simulated* interventions that assume a *particular state of the world* (note that interventions do not require any assumptions about the state of the world). For instance, answering a counterfactual question such as "*Would John have bought the chocolate last Friday had he seen the ad last week*?" requires us to know many things about John (and his environment) in the past. On the other hand, we don't need to know anything about John's life or environment from last week in order to perform an intervention (show John an ad and see whether he buys).

Contrary to interventions, counterfactuals can never be observed.

We can also view the difference between counterfactuals and interventions from another angle – two different counterfactual causal models can lead to the same interventional distribution.

For instance, Pearl (2009) describes a two-model scenario where the average (interventional) causal effect for a drug is equal to 0. This is true for both models.

The difference is that in one model, no patient is affected by the drug, while in the second model, all patients are affected by the drug – a pretty striking difference. As you can expect, counterfactual outcomes for both models differ (for more details, check out Pearl (2009, pp. 33–38)).

One idea that emphasizes the fundamental difference between counterfactuals and interventions is that interventional queries can be computed as the expected value of counterfactual queries over the population (Huszár, 2019). This fact further underlines the asymmetry between the two and, at the same time, reveals the deep, beautiful structure of Pearlian causal formalism.

Let's get weird (but formal)!

Now, it's time to express counterfactuals mathematically. Before we do so, let's set up an example so that we can refer to something familiar, while exploring the inherent weirdness of *the Counterfactual Wonderland*. Imagine you had a coffee this morning and now you feel bad in your stomach. Would you feel the same or better if you hadn't had your coffee?

Note that we cannot answer this question with interventions. A randomized experiment would only allow us to answer questions such as "*What is the probability that people similar to you react to coffee the way you reacted, given similar circumstances?*" or "*What is the probability that you'll feel bad after drinking a coffee in the morning on a day similar to today, given similar circumstances?*"

Counterfactuals are trying to answer a different question: "*Given an alternative world that is identical to ours and only differs in the fact that you did not drink coffee this morning (plus the necessary consequences of not drinking it), what is the probability that you'd feel bad in your stomach?*"

Let's try to formalize it. We'll denote the fact that you drank coffee this morning by $X = 1$ and the fact that you now feel bad as $Y_{X=1} = 1$. The $\cdot_{X=1}$ subscript informs us that the outcome, Y, happened in the world where you had your coffee in the morning ($X = 1$). The quantity we want to estimate, therefore, is the following:

$$P\left(Y_{X=0} = 1 \middle| X = 1, Y_{X=1} = 1\right)$$

We read this as the probability that you'd feel bad if you hadn't had your coffee, given you had your coffee and you feel bad.

Let's unpack it:

- $P\left(Y_{X=0} = 1\right)$ stands for the probability that you'd feel bad (in the alternative world) if you hadn't had your coffee
- $X = 1$ denotes that you had your coffee in the real world
- $Y_{X=1} = 1$ says that you had your coffee and you felt bad (in the real world)

Note that everything on the right side of the conditioning bar comes from the actual observation. The expression on the left side of the conditional bar refers to the alternative, hypothetical world.

Many people feel uncomfortable seeing this notation for the first time. The fact that we're conditioning on $X = 1$ to estimate the quantity in the world where $X = 0$ seems pretty counterintuitive. On a deeper level, this makes sense though. This notation makes it virtually impossible to reduce counterfactuals to *do* expressions (Pearl, Glymour, and Jewell, 2016). This reflects the inherent relationship between interventions and counterfactuals that we discussed earlier in this section.

Counterfactual notation

In this book, we follow the notation used by Pearl, Glymour, and Jewell (2016). There are also other options available in the literature. For instance, Peters, Janzing, and Schölkopf propose (2017) using different style of notation. Another popular choice is notation related to Donald Rubin's *potential outcomes* framework. It is worth making sure you understand the notation well. Otherwise, it can be a major source of confusion.

The fundamental problem of causal inference

Counterfactuals make it very clear that there's an inherent and fundamental problem with causal inference. Namely, we can never observe the same object (or subject) receiving two mutually exclusive treatments at the same time in the same circumstances.

You may wonder, given the fundamental problem of causal inference (Holland, 1986), whether there is any way to compute counterfactuals. It turns out that there is.

Computing counterfactuals

The basic idea behind computing counterfactuals is simple in theory, but the goal is not always easily achievable in practice. This is because computing counterfactuals requires an SCM that is *fully specified* at least for all the relevant variables. What does it mean? It means that we need to have full knowledge about functions that relate to relevant variables in the SCM and full knowledge about the values of *all* relevant exogenous variables in a system. Fortunately, if we know structural equations, we can compute noise variables describing the subject in the **abduction** step (see the following *Computing counterfactuals step by step* callout).

Computing counterfactuals step by step

Judea Pearl and colleagues (Pearl, Glymour and Jewell, 2016) proposed a three-step framework for computing counterfactuals:

- **Abduction**: Using evidence to calculate values of exogenous variables
- **Modification (originally called an action)**: Replacing the structural equation for the treatment with a counterfactual value
- **Prediction**: Using the modified SCM to compute the new value of the outcome under the counterfactual

We will apply this framework soon.

Let's take our coffee example. Let T denote our treatment – drinking coffee, while U will characterize you fully as an individual in our simplified world. $U = 1$ stands for coffee sensitivity, while $U = 0$ stands for a lack thereof. Additionally, let's assume that we know the causal mechanism for reaction to coffee. The mechanism is defined by the following SCM:

$$T := t$$

$$Y := TU + (T - 1)(U - 1)$$

Great! We know the outcome under the actual treatment, $Y_{T=1} = 1$ (you drank the coffee and you felt bad), but we don't know your characteristics (U). Can we do something about it?

It turns out that our model allows us to unambiguously deduct the value of U (that we'll denote as u), given we know the values of Y and T. Let's solve for u by transforming our structural equation for Y:

$$u = \frac{T + Y - 1}{2T - 1}$$

Now, let's assign the values for T and Y:

$$u = \frac{1 + 1 - 1}{2 * 1 - 1} = \frac{2 - 1}{2 - 1} = 1$$

The value we obtained for U reveals that you're a coffee-sensitive person.

This step (solving for the values of exogenous variables U) is called **abduction**.

Now, we have all the elements necessary to compute counterfactual outcomes at our disposal – a causal model and knowledge about your personal characteristics.

We're ready for the next step, **modification**. We will fix the value of our treatment at the counterfactual of interest, ($T = 0$):

$$T := 0$$

$$Y := 0U + (0 - 1)(U - 1)$$

Finally, we're ready for the last step, **prediction**. To make a prediction, we need to substitute U with the value(s) of your personal characteristic(s) that we computed before:

$$Y := 0 * 1 + (0 - 1)(1 - 1) = 0$$

And here is the answer – you wouldn't feel bad if you hadn't had your coffee!

Neat, isn't it?

Deterministic and probabilistic counterfactuals

You might have noticed that when we introduced our coffee example, we were speaking in terms of probability, although we solved it deterministically. That's a great observation!

In the real world, we are not always able to compute counterfactuals deterministically. Fortunately, the three-step framework that we introduced earlier generalizes well for probabilistic settings. To learn more on how to compute probabilistic counterfactuals, please refer to *Chapter 4* of *Causal inference in statistics: A primer* (Pearl, Glymour, and Jewell, 2016) or an excellent YouTube video by Brady Neal (Neal, 2020).

Time to code!

The last thing we'll do before concluding this section is to implement our counterfactual computation in Python:

1. First, we'll create a `CounterfactualSCM` class with three methods corresponding to the three steps of counterfactual computation – *abduction*, *modification*, and *prediction*:

```
class CounterfactualSCM:

    def abduct(self, t, y):
        return (t + y - 1)/(2*t - 1)

    def modify(self, t):
        return lambda u: t * u + (t - 1) * (u - 1)

    def predict(self, u, t):
        return self.modify(t)(u)
```

   Note that each method implements the steps that we performed in the preceding code:

   - `.abduct()` computes the value of U, given the values for treatment and the actual outcome
   - `.modify()` modifies the SCM by assigning t to T
   - `.predict()` takes the modified SCM and generates the counterfactual prediction by assigning the actual value of U to the modified SCM

2. Let's instantiate the SCM and assign the known treatment and outcome values to the `t` and `y` variables respectively:

```
coffee = CounterfactualSCM()
t = 1
y = 1
```

3. Next, let's obtain the value of U by performing *abduction* and print out the result:

```
u = coffee.abduct(t=t, y=y)
u
```

   The result is congruent with our computations:

```
1.0
```

4. Finally, let's compute the counterfactual prediction:

```
coffee.predict(u=u, t=0)
```

   Et voila! Here's our answer:

```
0.0
```

If you hadn't had your coffee in the morning, you'd feel better now.

This concludes our section on counterfactuals. We learned that counterfactuals are different than interventions. At the same time, there's a deeper link between rung two and rung three of *the Ladder of Causation*. We learned about the fundamental problem of causal inference, and we've seen that even though it's hard, we can compute counterfactuals when we meet certain assumptions. Finally, we practiced computing counterfactuals using mathematical tools and Python code.

Counterfactuals will briefly return in *Part 2*, *Causal Inference*, where we'll see them in the context of counterfactual explanations.

Extra – is all machine learning causally the same?

So far, when we have spoken about machine learning, we mainly mean supervised methods. You might wonder what the relationship is between other types of machine learning and causality.

Causality and reinforcement learning

For many people, the first family of machine learning methods that come to mind when thinking about causality is **reinforcement learning** (**RL**).

In the classic formulation of RL, an agent interacts with the environment. This suggests that an RL agent can make interventions in the environment. Intuitively, this possibility moves RL from an *associative* rung one to an *interventional* rung two. Bottou et al. (2013) amplify this intuition by proposing that causal models can be reduced to multiarmed bandit problems – in other words, that RL bandit algorithms are special cases of rung two causal models.

Although the idea that all RL is causal might seem intuitive at first, the reality is more nuanced. It turns out that even for certain bandit problems, the results might not be optimal if we do not model causality explicitly (Lee and Bareinboim, 2018).

Moreover, *model-based RL algorithms* can suffer from confounding. This includes models such as the famous DeepMind's MuZero (Schrittwieser et al., 2019), which was able to master games such as Chess, Go, and many others without explicitly knowing the rules.

Rezende and colleagues (2020) proposed to deconfound these models by using **back-door criterion** (we will learn more about back-door criterion in *Chapter 6*. Wang et al. (2022) proposed adding a causal model to the RL algorithm and using a ratio between causal and non-causal terms in a reward function, in order to encourage an agent to explore the space where the non-causal term is larger, which suggests that the causal model might be causally biased.

For a comprehensive overview of the literature on the relationships between causality and RL, check Kaddour et al. (2022), pp. 70–98.

Causality and semi-supervised and unsupervised learning

Peters et al. (2017, pp. 72–74) proposed that semi-supervised methods can be used for causal discovery, leveraging the information-theoretic asymmetry between conditional and unconditional probabilities. Based on a similar idea, Sgouritsa et al. (2015) proposed *unsupervised inverse regression* as a method to discover which causal direction between two variables is the correct one. Wu et al. (2022) explored the links between causality and semi-supervised learning in the context of *domain adaptation*, while Vowels et al. (2021) used unsupervised neural embeddings for the causal discovery of video representations of *dynamical systems*.

Furthermore, data augmentation in semi-supervised learning can be seen as a form of representation disentanglement, making the learned representations *less* confounded. As deconfounding is only partial in this case, these models cannot be considered fully causal, but perhaps we could consider putting these partially deconfounded representations somewhere between rungs one and two of *the Ladder of Causation* (Berrevoets, 2023).

In summary, the relationship between different branches of contemporary machine learning and causality is nuanced. That said, most broadly adopted machine learning models operate on rung one, not having a causal world model. This also applies to large language models such as GPT-3, GPT-4, LlaMA, or LaMDA, and other popular generative models such as DALL-E 2. Models such as GPT-4 can sometimes correctly answer causal or counterfactual queries, yet their general performance suggests that these abilities do not always generalize well (for a brief discussion and references check *Chapter 11*).

Wrapping it up

In this chapter, we introduced the concept of *the Ladder of Causation*. We discussed each of the three rungs of the ladder: associations, interventions, and counterfactuals. We presented mathematical apparatus to describe each of the rungs and translated the ideas behind them into code. These ideas are foundational for causal thinking and will allow us to understand more complex topics further on in the book.

Additionally, we broadened our perspective on causality by discussing the relationships between causality and various families of machine learning algorithms.

In the next chapter, we'll take a look at the link between observations, interventions, and linear regression to see the differences between rung one and rung two from yet another perspective. Ready?